Sequences & Series

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 4x + 4$ in box plots and measures of spread, evaluate $f(5)$.

Final Answer: 24. Substitute $x = 5$: $f(5) = 4(5) + 4 = 24$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

For the model $f(x) = 5x + 5$ in data distributions, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 5(2) + 5 = 15$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A data set recorded for data distributions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 20.5. The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

4 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 6$ in comparisons of data sets, evaluate $f(2)$.

Final Answer: 18. Substitute $x = 2$: $f(2) = 6(2) + 6 = 18$.

5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A data set recorded for comparisons of data sets consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

6 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

For the model $f(x) = 2x + 7$ in two-way frequency tables, evaluate $f(2)$.

Final Answer: 11. Substitute $x = 2$: $f(2) = 2(2) + 7 = 11$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for two-way frequency tables consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In an arithmetic sequence, the 4th term is 17 and the 10th term is 47. Find the first term a_1, the common difference d, and the sum of the first 25 terms S_{25}.

Using $a_n = a_1 + (n-1)d$: $a_4 = a_1 + 3d = 17,\ a_{10} = a_1 + 9d = 47.$ Subtract the first equation from the second: $6d = 30 \Rightarrow d = 5.$ Substitute $d = 5$ into the first equation: $a_1 + 3(5) = 17 \Rightarrow a_1 = 17 - 15 = 2.$ Find the sum of the first 25 terms using $S_n = \frac{n}{2}[2a_1 + (n-1)d]$:

$S_{25} = \frac{25}{2}[2(2) + 24(5)] = \frac{25}{2}[4 + 120] = \frac{25}{2}(124) = 25 \times 62 = 1550.$

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Find the coefficient of x^2 in the expansion of $(3 - 2x)^5$.

$\binom{5}{2}(3)^3(-2x)^2 = 10 \times 27 \times 4x^2 = 1080x^2$. Coefficient: 1080.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

Consider the ordered dataset of 12 numbers: 14, 16, 18, 20, 22, 24, 26, 28, 30, 34, 38, 72. (a) Determine Q_1, the median, Q_3, and the IQR. (b) Use Tukey's $1.5 \times \text{IQR}$ rule to determine whether 72 is an outlier.

(a) Five-Number Summary Calculations:Lower half (first 6 values): 14, 16, 18, 20, 22, 24 $\Rightarrow Q_1 = \frac{18 + 20}{2} = \mathbf{19}$.Median of all 12 values: $\frac{24 + 26}{2} = \mathbf{25}$.Upper half (last 6 values): 26, 28, 30, 34, 38, 72 $\Rightarrow Q_3 = \frac{30 + 34}{2} = \mathbf{32}$.IQR $= Q_3 - Q_1 = 32 - 19 = \mathbf{13}$. (b) Outlier Boundary Check:

Upper Outlier Boundary $= Q_3 + 1.5 \times \text{IQR} = 32 + 1.5(13) = 32 + 19.5 = \mathbf{51.5}$. Since $72 > 51.5$, the va.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

An equilateral triangle has a side length of 12 cm. Find: (a) the exact length of its altitude, and (b) its exact area.

(a) Altitude using $30^\circ - 60^\circ - 90^\circ$ triangle: Dropping an altitude splits the equilateral triangle into two congruent $30^\circ - 60^\circ - 90^\circ$ right triangles with short leg $b/2 = 6$ cm and hypotenuse 12 cm. The altitude is opposite the 60° angle: $h = 6\sqrt{3}$ cm. (b) Area formula: $A = \frac{1}{2}bh = \frac{1}{2}(12)(6\sqrt{3}) = \mathbf{36\sqrt{3}}$ cm². (Alternatively.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Solve $2x^2 - 6x + 5 = 0$ by completing the square, expressing all solutions in exact $a \pm bi$ form.

Divide by the leading coefficient 2 or factor it out from the variable terms: $2(x^2 - 3x) = -5 \Rightarrow x^2 - 3x = -\frac{5}{2}.$ Add $\left(-\frac{3}{2}\right)^2 = \frac{9}{4}$ to both sides: $x^2 - 3x + \frac{9}{4} = -\frac{5}{2} + \frac{9}{4} = -\frac{10}{4} + \frac{9}{4} = -\frac{1}{4}.$ Write as a perfect square and take the square root of both sides: $\left(x - \frac{3}{2}\right)^2 = -\frac{1}{4} \Rightarrow x - \frac{3}{2} = \pm\sqrt{-\frac{1}{4}} = \pm\frac{i}{2}.$ $x = \frac{3}{2} \pm \frac{1}{2}i.$

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two assembly line components operate independently. Component A has a 95% reliability rate (probability of functioning properly is 0.95), and Component B has a 90% reliability rate (probability of functioning is 0.90). What is the probability that at least one of the two components fails?

Step 1: Use the complement rule. The event "at least one fails" is the exact complement of "both components function properly": $P(\geq 1 \text{ fails}) = 1 - P(\text{both function})$. Step 2: Since the components operate independently: $P(\text{both function}) = P(A) \times P(B) = 0.95 \times 0.90 = 0.855$. Step 3: Calculate the complement: $P(\geq 1 \text{ fails}) = 1 - 0.855 = \mathbf{0.145\ (14.5\%)}$. The probability that at least one component fails is 0.145 (or 14.5%).

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For what value of the constant a will the system of equations $ax + 4y = 7$ and $3x + 2y = 5$ have no solution?

Step 1: A linear system has no solution when the lines are parallel and distinct (slopes are equal, y-intercepts differ). In standard form, this requires: $\frac{a}{3} = \frac{4}{2} eq \frac{7}{5}$ Step 2: Solve the slope equality: $\frac{a}{3} = 2 \Rightarrow a = 2 \times 3 = \mathbf{6}$ Check: When $a = 6$, the ratio of coefficients is $\frac{6}{3} = \frac{4}{2} = 2$, while the constant ratio is $\frac{7}{5} = 1.4 eq 2$. Therefore, the lines are strictly parallel with no solution.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Statistics & Data Analysis

Find the equation of the perpendicular bisector of the line segment joining $P(2, -1)$ and $Q(8, 7)$. Give your answer in the form $y = mx + c$.

Step 1: Find the midpoint M of PQ: $M = \left(\frac{2+8}{2}, \frac{-1+7}{2}\right) = (5, 3)$ Step 2: Find the gradient of line PQ: $m_{PQ} = \frac{7-(-1)}{8-2} = \frac{8}{6} = \frac{4}{3}$ Step 3: Determine the perpendicular gradient: $m_{\perp} = -\frac{3}{4}$ Step 4: Form the equation passing through $M(5, 3)$: $y - 3 = -\frac{3}{4}(x - 5) \Rightarrow y - 3 = -\frac{3}{4}x + \frac{15}{4}$
$y = -\frac{3}{4}x + \frac{27}{4}$ or $y = -0.75x + 6.75$

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Quadratic Functions

Convert $\frac{7\pi}{6}$ radians to degrees, and find the exact coordinates of its terminal point on the unit circle.

Step 1: Convert to degrees: $\theta = \frac{7\pi}{6} \times \frac{180^\circ}{\pi} = 7 \times 30^\circ = \mathbf{210^\circ}$. Step 2: Terminal coordinates in Quadrant III: The reference angle is $210^\circ - 180^\circ = 30^\circ$. In Quadrant III, both coordinates are negative: $x = \cos(210^\circ) = -\cos(30^\circ) = -\frac{\sqrt{3}}{2}, \quad y = \sin(210^\circ) = -\sin(30^\circ) = -\frac{1}{2}$. The terminal point is $\left(\mathbf{-\frac{\sqrt{3}}{2}, -\frac{1}{2}}\right)$.

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17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 3.01 Exponents & Polynomials**

A rectangular water storage reservoir has dimensions 2.5 meters by 3 meters by 1.2 meters. A drainage pump empties the reservoir at a constant rate of 75 liters per minute. Given that $1\ \text{m}^3 = 1000\ \text{liters}$, how many hours will it take to completely empty the full reservoir?

Step 1: Find the reservoir volume in cubic meters: $V = 2.5 \times 3.0 \times 1.2 = 9.0\ \text{m}^3$. Step 2: Convert volume to liters: $V = 9.0\ \text{m}^3 \times 1000\ \text{L/m}^3 = 9000\ \text{liters}$. Step 3: Compute drainage time in minutes: $t_{\text{min}} = \frac{9000\ \text{L}}{75\ \text{L/min}} = 120\ \text{minutes}$. Step 4: Convert to hours: $t = \frac{120\ \text{min}}{60\ \text{min/h}} = \mathbf{2}\ \text{hours}$. It takes exactly 2 hours.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 4.01 Functions & Transformations**

In the system $ax + 4y = 18$ and $3x + 2y = 9$, the system has no solution. What is the value of a?

A system of linear equations has no solution if the lines are parallel (equal slopes) with different y-intercepts. Equal slopes require the ratio of the x-coefficients to equal the ratio of the y-coefficients: $\frac{a}{3} = \frac{4}{2} = 2$. Multiplying both sides by 3 gives $a = 6$. (Since $\frac{18}{9} = 2$, note that the constants would produce identical lines if $a = 6$; however, to make lines parallel and distinct with non-overlapping constraints, $a = 6$).

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 5.01 Exponential & Logarithmic Functions**

A line segment connects points $P(-3, 7)$ and $Q(5, -1)$. Determine: (a) the exact length of PQ, (b) the coordinates of the midpoint M, and (c) the slope of the perpendicular bisector of PQ.

(a) Distance formula: $PQ = \sqrt{(5-(-3))^2 + (-1-7)^2} = \sqrt{8^2 + (-8)^2} = \sqrt{64+64} = \sqrt{128} = \mathbf{8\sqrt{2}}$. (b) Midpoint formula: $M = \left(\frac{-3+5}{2}, \frac{7+(-1)}{2}\right) = \left(\frac{2}{2}, \frac{6}{2}\right) = \mathbf{(1, 3)}$. (c) Perpendicular slope: Slope of PQ is $m = \frac{-1-7}{5-(-3)} = \frac{-8}{8} = -1$. The perpendicular bisector has negative reciprocal slope: $m_{\perp} = -\frac{1}{-1} = \mathbf{+1}$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 6.01 Systems & Inequalities**

An investment fund gains 30% in value during Year 1, decreases by 20% in Year 2, and increases by 10% in Year 3. Determine the overall net percent change of the fund across the three-year period.

Step 1: Express each yearly change as a decimal multiplier:Year 1 (+30%): $1 + 0.30 = 1.30$Year 2 (-20%): $1 - 0.20 = 0.80$Year 3 (+10%): $1 + 0.10 = 1.10$ Step 2: Multiply the factors sequentially: $M_{\text{net}} = 1.30 \times 0.80 \times 1.10 = 1.04 \times 1.10 = 1.144$. Step 3: Convert to net percentage change: $\text{Net Percent Change} = (1.144 - 1) \times 100\% = \mathbf{+14.4\%}$. The investment gained an overall 14.4%.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 9$ in frequency data displays, evaluate $f(2)$.

a. 13 ✓

b. 15

c. 4

d. 17

Substitute $x = 2$: $f(2) = 2(2) + 9 = 13$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 3$ in measures of center, evaluate $f(2)$.

a. 9 ✓

b. 12

c. 6

d. 13

Substitute $x = 2$: $f(2) = 3(2) + 3 = 9$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for frequency data displays consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for measures of center consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Prove by induction that $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$. Step: assume true for n, verify for $n+1$.

a. Not provable by induction

b. Only for $n \leq 100$

c. $\frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$. ✓✓

d. Only for even n

Applying Sequences & Series principles, Standard induction step. Therefore, the verified correct option is $\frac{n(n+1)}{2} + (n+1) = \frac{(n+1)(n+2)}{2}$. ✓.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Prove: in any geometric sequence with positive terms, the geometric mean of two terms equals the intermediate terms.

a. Not always true

b. Only for $r = 1$

c. Use AM-GM

d. **If** a, ar, ar^2**:** $\sqrt{a \cdot ar^2} = ar =$ **middle term** ✓

$\sqrt{a \cdot ar^2} = a\sqrt{r^2} = ar$. ✓.

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27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $(a_n)_{n \geq 1}$ be an arithmetic sequence of real numbers with first term $a_1 > 0$ and non-zero common difference d. The terms a_1, a_3, and a_4 form the first three terms of a convergent geometric sequence $(g_k)_{k \geq 1}$, such that $g_1 = a_1$, $g_2 = a_3$, and $g_3 = a_4$. Let $S_\infty = \sum_{k=1}^{\infty} g_k$ denote the sum to infinity of the geometric sequence, and let $S_n = \sum_{j=1}^{n} a_j$ denote the n-th partial sum of the arithmetic sequence. If $M = \max_{n \geq 1} S_n$ is the maximum value attained by the partial sums of the arithmetic sequence, what is the value of $\frac{M}{S_\infty}$?

a. $\frac{5}{4}$ ✓

b. $\frac{81}{64}$

c. $\frac{9}{8}$

d. $\frac{5}{8}$

1. **Determine the relation between d and a_1:** The terms of the arithmetic sequence are $a_n = a_1 + (n-1)d$. Therefore: $g_1 = a_1, \quad g_2 = a_3 = a_1 + 2d, \quad g_3 = a_4 = a_1 + 3d$ Since g_1, g_2, g_3 form a geometric progression, the geometric mean property requires: $g_2^2 = g_1 g_3 \Rightarrow (a_1 + 2d)^2 = a_1(a_1 + 3d)$ Expanding both sides: $a_1^2 + 4a_1 d + 4d^2 = a_1^2 + 3a_1 d \; a_1 d + 4d^2 = 0 \Rightarrow d(a_1 + 4d) = 0$ Since $de0$, we have $a_1 + 4d = 0$, which gives.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider an infinite geometric series with first term a and common ratio r, where $|r| < 1$: $a + ar + ar^2 + ar^3 + \ldots = 12$ If each term in the series is squared, the sum of the resulting infinite geometric series is: $a^2 + a^2 r^2 + a^2 r^4 + a^2 r^6 + \ldots = 72$ What is the value of the common ratio, r?

a. $\frac{1}{3}$ ✓

b. $\frac{1}{2}$

c. $\frac{2}{3}$

d. $\frac{1}{4}$

Step 1: Use the sum formula for an infinite geometric series with $|r| < 1$: $S = \frac{a}{1-r} = 12 \Rightarrow a = 12(1-r)$ Step 2: When each term is squared, the new series has first term a^2 and common ratio r^2. Since $|r| < 1$, we also have $r^2 < 1$. Its sum is: $S_{\text{squared}} = \frac{a^2}{1-r^2} = 72$ Step 3: Factor the denominator using the difference of squares, $1 - r^2 = (1-r)(1+r)$: $\frac{a^2}{(1-r)(1+r)} = \left(\frac{a}{1-r}\right)\left(\frac{a}{1+r}\right) = 72$ Substi.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A geometric sequence has its first term a_1 and common ratio r. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. Find the value of a_1 given that $req1$.

a. 1

b. **3** ✓

c. 7

d. 9

Let the geometric sequence be $a_1, a_1r, a_1r^2, \ldots$. The sum of the first n terms of a geometric sequence is given by $S_n = \frac{a_1(1-r^n)}{1-r}$ (for $req1$). We are given that the sum of the first 3 terms is 21: $S_3 = \frac{a_1(1-r^3)}{1-r} = 21$ (Equation 1) We are also given that the sum of the first 6 terms is 189: $S_6 = \frac{a_1(1-r^6)}{1-r} = 189$ (Equation 2) We can rewrite Equation 2 using the difference of squares factorization, $1 - r^6 = (1 - r^3)(1 + r^3)$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In a geometric sequence of positive real numbers, the sum of the first two terms is 18, and the sum of the first four terms is 90. What is the fifth term of the sequence?

a. 48

b. 72

c. 96 ✓

d. 192

Let a_1 be the first term and r be the common ratio of the geometric sequence. Since all terms are positive real numbers, $a_1 > 0$ and $r > 0$. The sum of the first two terms is: $S_2 = a_1 + a_1r = a_1(1 + r) = 18$ The sum of the first four terms is: $S_4 = a_1 + a_1r + a_1r^2 + a_1r^3 = a_1(1 + r)(1 + r^2) = 90$ Substituting $a_1(1 + r) = 18$ into the equation for S_4: $18(1 + r^2) = 90$ $1 + r^2 = 5 \Rightarrow r^2 = 4$ Since $r > 0$, we have $r = 2$. Now substitute $r = 2$ back to solve for a_1.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the sequence a_n defined by $a_1 = 3$ and $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$ for $n \geq 1$. If the sequence converges to a limit L, find the value of $L^2 - 2L$.

a. **1** ✓

b. 2

c. 3

d. 4

The sequence is defined by the recurrence relation $a_{n+1} = \frac{a_n^2 + 4}{2a_n}$. If the sequence converges to a limit L, then as $n \to \infty$, we have $a_{n+1} \to L$ and $a_n \to L$. Substituting L into the recurrence relation, we get: $L = \frac{L^2 + 4}{2L}$ Multiplying both sides by $2L$ (assuming $Leq0$), we have: $2L^2 = L^2 + 4$ $L^2 = 4$ This gives $L = 2$ or $L = -2$. Let's check the first few terms of the sequence to determine the sign of the limit. $a_1 = 3$..

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

An arithmetic sequence begins with the terms: $8, 13, 18, 23, \ldots$ What is the 15th term of the sequence?

a. 78 ✓

b. 83

c. 73

d. 117

To find the 15th term of an arithmetic sequence, use the explicit formula for the nth term: $a_n = a_1 + (n-1)d$ 1. Identify the first term: $a_1 = 8$. 2. Find the common difference d by subtracting consecutive terms: $d = 13 - 8 = 5$ 3. Substitute $n = 15$, $a_1 = 8$, and $d = 5$ into the formula: $a_{15} = 8 + (15-1)(5) = 8 + 14(5) = 8 + 70 = 78$ Common Misconceptions: - Choice B (83) results from forgetting to subtract 1 from n, computing $8 + 15(5) = 83$. - Choice C (73) results from off-by-one in.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

An arithmetic sequence has a first term of $a_1 = 5$ and a common difference of $d = 3$. What is the value of the 6th term, a_6?

a. **20** ✓

b. 23

c. 17

d. 18

The general term of an arithmetic sequence is given by the formula: $a_n = a_1 + (n-1)d$ Substitute $a_1 = 5$, $d = 3$, and $n = 6$: $a_6 = 5 + (6-1) \cdot 3 = 5 + 5 \cdot 3 = 5 + 15 = 20$ Misconception Analysis: - 23 comes from using $a_1 + n \cdot d = 5 + 6(3) = 23$, forgetting to subtract 1 from n. - 17 is a_5, resulting from an off-by-one counting error. - 18 comes from directly multiplying $n \cdot d = 6 \cdot 3 = 18$, neglecting the initial term.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the arithmetic sequence where the first term $a_1 = 7$ and the common difference $d = -3$. Which of the following is the explicit formula for the nth term of this sequence?

a. $a_n = 7 + 3(n-1)$

b. $a_n = 7 - 3(n-1)$ ✓

c. $a_n = 3 - 7(n-1)$

d. $a_n = -3 + 7(n-1)$

An arithmetic sequence is defined by the formula $a_n = a_1 + (n-1)d$, where a_n is the nth term, a_1 is the first term, and d is the common difference. In this problem, we are given that the first term $a_1 = 7$ and the common difference $d = -3$. Substituting these values into the formula, we get $a_n = 7 + (n-1)(-3)$, which can be rewritten as $a_n = 7 - 3(n-1)$.

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A sequence is defined by the recurrence relation $a_n = 2a_{n-1} - a_{n-2} + 3$ for $n \geq 3$. If the first term $a_1 = 5$ and the second term $a_2 = 8$, find the value of a_4.

a. 17

b. 20

c. **23** ✓

d. 26

The recurrence relation is given by $a_n = 2a_{n-1} - a_{n-2} + 3$ for $n \geq 3$. We are given $a_1 = 5$ and $a_2 = 8$. To find a_3, we set $n = 3$: $a_3 = 2a_2 - a_1 + 3$ $a_3 = 2(8) - 5 + 3$ $a_3 = 16 - 5 + 3$ $a_3 = 11 + 3$ $a_3 = 14$ To find a_4, we set $n = 4$: $a_4 = 2a_3 - a_2 + 3$ $a_4 = 2(14) - 8 + 3$ $a_4 = 28 - 8 + 3$ $a_4 = 20 + 3$ $a_4 = 23$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $|r| < 1$, prove $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$ using the limit of partial sums.

a. Use calculus only

b. Cannot be proved

c. $S_N = \frac{1-r^{N+1}}{1-r} \to \frac{1}{1-r}$ **as** $r^{N+1} \to 0$ ✓

d. Only for $0 < r < 1$

Applying Sequences & Series principles, Classic geometric series proof. Therefore, the verified correct option is $S_N = \frac{1-r^{N+1}}{1-r} \to \frac{1}{1-r}$ as $r^{N+1} \to 0$.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Prove $\sum_{k=1}^{n} k^3 = \left(\frac{n(n+1)}{2}\right)^2$ using the formula for arithmetic sums.

a. Cannot be proved simply

b. By FTA

c. **By induction: base** $n = 1$ **trivially true; induction step involves algebraic manipulation** ✓

d. Use telescoping

Applying Sequences & Series principles, Standard induction. Therefore, the verified correct option is By induction: base $n = 1$ trivially true; induction step involves algebraic manipulation.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Functions & Transformations

Find a_1 and d for the arithmetic sequence where $S_{10} = 155$ and $a_{10} = 28$.

a. $a_1 = 3,\ d = \frac{25}{9}$ ✓

b. $a_1 = 5,\ d = 3$

c. $a_1 = 1,\ d = 3$

d. $a_1 = 3,\ d = 3$

From $S_{10} = \frac{10}{2}(a_1 + a_{10}) = 155$ with $a_{10} = 28$, we get $5(a_1 + 28) = 155 \Rightarrow a_1 = 3$. Then $a_{10} = a_1 + 9d \Rightarrow 28 = 3 + 9d \Rightarrow d = \frac{25}{9}$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Prove by induction: $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$.

a. Not provable

b. Only for even n

c. Use calculus

d. **Base:** $n = 1$**:** $1 = 6/6$**. ✓. Step: add** $(n+1)^2$ **and factor.** ✓

Applying Sequences & Series principles, Standard induction. Therefore, the verified correct option is Base: $n = 1$: $1 = 6/6$. ✓. Step: add $(n+1)^2$ and factor..

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.01 Systems & Inequalities

Find the closed form for F_n (Fibonacci) using Binet's formula.

a. $F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$ **where** $= 1 + 52,\ = 1 - 52$ ✓

b. $F_n = \phi^{n-1}$

c. $F_n = 2^n/5$

d. $F_n = n\phi$

Applying Sequences & Series principles, Binet's formula. Therefore, the verified correct option is $F_n = \frac{\phi^n - \psi^n}{\sqrt{5}}$ where $= 1 + 52,\ = 1 - 52$.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Factoring Polynomials

Using partial fractions, find $\sum_{k=1}^{n} \frac{1}{(2k-1)(2k+1)}$.

a. 1

b. $\frac{1}{2}$

c. $\frac{n}{2n-1}$

d. $\frac{n}{2n+1}$ ✓

PFD: $\frac{1}{2}\left(\frac{1}{2k-1} - \frac{1}{2k+1}\right)$. Telescopes to $\frac{1}{2}\left(1 - \frac{1}{2n+1}\right) = \frac{n}{2n+1}$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Radicals & Exponents

The formula for the sum of n terms of an arithmetic series is:

a. $S_n = \frac{a_1}{1-r}$

b. $S_n = \frac{n}{2}(a_1 + a_n)$ ✓

c. $S_n = na_1$

d. $S_n = a_1 \cdot \frac{r^n - 1}{r - 1}$

Applying Sequences & Series principles, Arithmetic sum formula. Therefore, the verified correct option is $S_n = \frac{n}{2}(a_1 + a_n)$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Linear Equations & Lines

A geometric series with $|r| < 1$ has an infinite sum of:

a. $S = a_1 \cdot r$

b. $S = \frac{a_1}{1-r}$ ✓

c. Undefined

d. $S = \frac{a_1}{r}$

Applying Sequences & Series principles, Infinite geometric series sum. Therefore, the verified correct option is $S = \frac{a_1}{1-r}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Rational Expressions

Find the first term and common difference of the AP whose pth term is $3p + 2$.

a. $a_1 = 2, d = 3$

b. $a_1 = 5, d = 3$ ✓

c. $a_1 = 5, d = 2$

d. $a_1 = 3, d = 2$

Applying Sequences & Series principles, $a_p = 3p + 2 \Rightarrow a_1 = 5, d = 3$. Therefore, the verified correct option is $a_1 = 5, d = 3$.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 11.01 Statistics & Data Analysis

Use sigma notation for $1^2 + 2^2 + 3^2 + \cdots + n^2$.

a. $\prod_{k=1}^{n} k^2$

b. $\sum_{k=0}^{n} k^2$

c. $\sum_{k=1}^{n} k^2$ ✓

d. n^2

Applying Sequences & Series principles, Standard sigma notation. Therefore, the verified correct option is $\sum_{k=1}^{n} k^2$.

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